



International University, VNU-HCMC

School of Computer Science and Engineering

Lecture 6 Data Visualization and Communication

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Purpose of the lecture

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- Enable students to communicate Big Data analytics and model results effectively using visualizations
- Distinguish between exploratory and explanatory visualization in analytics workflows
- Select appropriate visualizations for models and performance metrics
- Design dashboards and visual narratives to support decision making
- Identify and avoid common visualization pitfalls in Big Data contexts



Warm-up question

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- A machine learning model achieves high accuracy. Why might this result still fail to support decision-making without proper visualization?
- Have you ever seen a chart or dashboard that confused you more than it helped? What do you think caused that problem?

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Outlines

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Visualization of Big Data Analytics Results

Exploratory vs. Explanatory Visualization

Visualizations for Models and Metrics

Dashboards for Analytics Communication

Data Storytelling for Decision Making

Common Visualization Pitfalls in Big Data

Applications & Case Studies

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Visualization of Big Data Analytics Results

Communicating insights effectively from complex data

- 1 **Purpose of visualizing analytics results**
- 2 **Types of analytics outputs to visualize**
- 3 **Interpreting Model Outputs through Visualization**
- 4 **Role of visualization**
- 5 **Challenges when visualizing Big Data results**

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Purpose of visualizing analytics results

- Transform complex analytical outputs into understandable insights
- Support interpretation and validation of model results
- Enable comparison between models, parameters, or scenarios
- Facilitate communication with non-technical stakeholders
- Assist decision-making based on data-driven evidence

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Visualization Design Principles

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- Match *question type* → *chart type* (compare / trend / distribution / relationship / composition)
- Reduce clutter (data-ink, avoid chartjunk)
- Label & annotate: title = message, not topic
- Choose scale carefully (zero baseline when needed)
- Color: consistent, limited, colorblind-safe

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Chart Selection Map

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Analytical Question	What you want to show	Recommended Chart	Example
Compare values	Which category is larger?	Bar chart	Sales by region
Trend over time	How does it change?	Line chart	Revenue by month
Distribution	How data is spread?	Histogram/ Box plot	Age distribution
Relationship	Do variables relate?	Scatter plot	Study time vs GPA
Composition	Parts of a whole	Stacked bar/ Pie	Market share

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Chart Selection Map

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Analytical Question	What you want to show	Recommended Chart	Example
Ranking	Top & bottom items	Sorted bar chart	Top 10 products
Correlation strength	Strength of relationships	Heatmap	Feature correlation
Pattern groups	Natural clusters	Cluster scatter	Customer segments
Outliers	Unusual values	Box plot/ anomaly plot	Fraud detection
Model performance	Model accuracy & errors	Confusion matrix/ ROC	Classification

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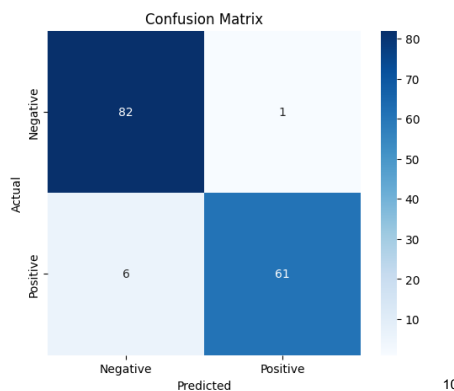
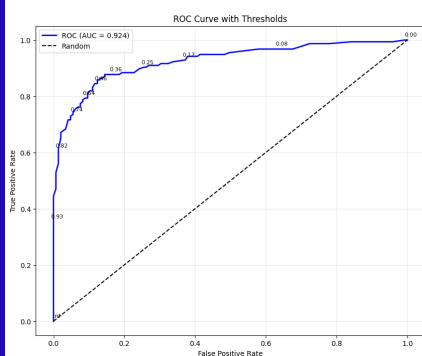


Types of analytics outputs to visualize

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Model Performance Metrics

- Classification: accuracy, precision, recall, F1-score, ROC curves
- Regression: error metrics, residual plots
- Purpose: evaluate and compare model effectiveness



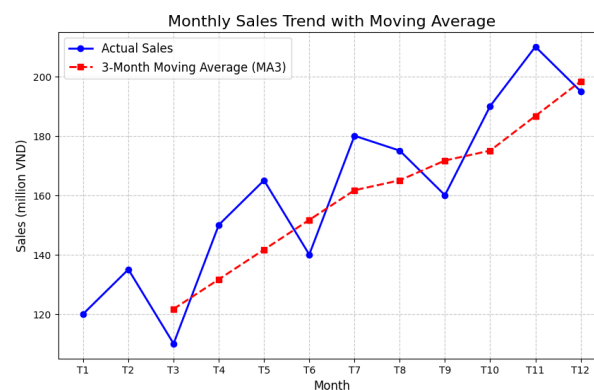
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Types of analytics outputs to visualize

Patterns and Trends

- Temporal trends and seasonality
- Correlations and relationships between variables
- Purpose: identify meaningful insights from large datasets



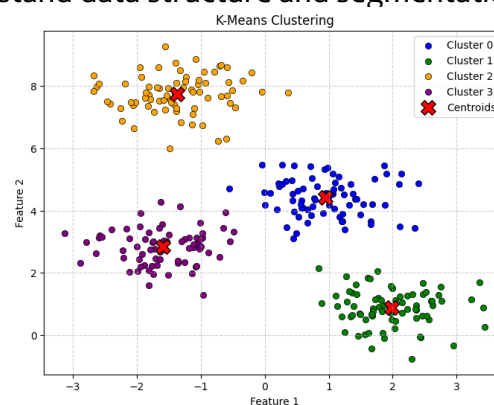
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Types of analytics outputs to visualize

Clustering and Grouping Results

- Cluster assignments and boundaries
- Cluster characteristics and sizes
- Purpose: understand data structure and segmentation



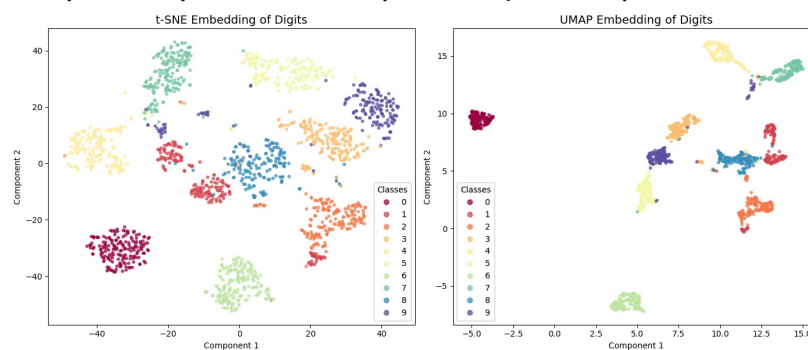
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Types of analytics outputs to visualize

High-Dimensional Data Representations

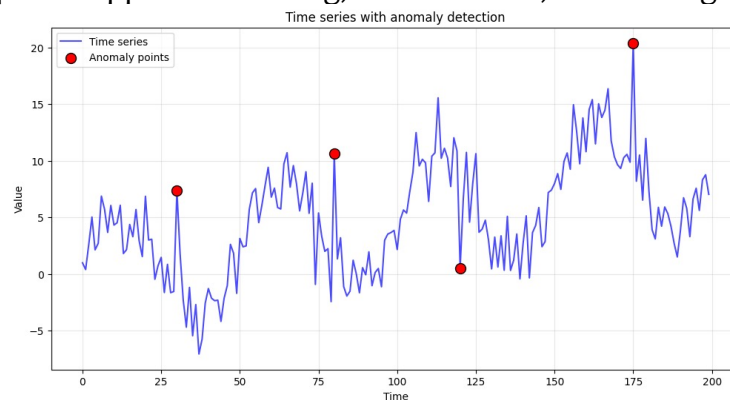
- Dimensionality reduction outputs (e.g., projections, embeddings)
- Feature distributions after transformation
- Purpose: improve the interpretability of complex data



Types of analytics outputs to visualize

Anomaly and Outlier Detection Results

- Detected anomalies or rare events
- Deviation from normal patterns
- Purpose: support monitoring, risk detection, and alerting





Interpreting Model Outputs through Visualization

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- Machine learning and Big Data models often produce abstract, numerical, or high-dimensional outputs that are difficult to interpret directly
- Visualization translates these outputs into visual patterns that align with human cognitive abilities

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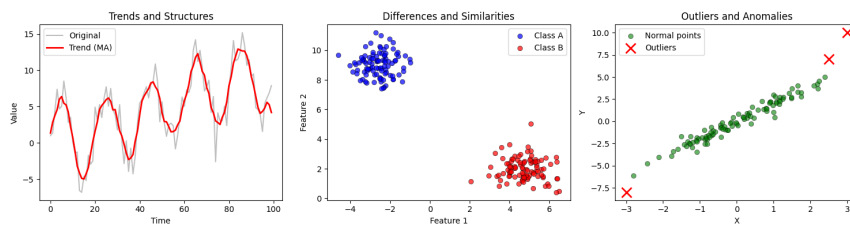


Interpreting Model Outputs through Visualization

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Helps humans recognize:

- Trends and structures
- Differences and similarities
- Outliers and anomalies



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Interpreting Model Outputs through Visualization

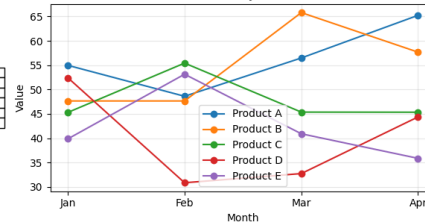
- Reduces the cognitive gap between:
 - What the model computes
 - What users need to understand
- Supports trust and confidence in analytics results by making model behavior more transparent

Bridging the Cognitive Gap

Raw Data (Difficult to interpret)

	Jan	Feb	Mar	Apr
Product A	55.0	48.6	56.5	65.2
Product B	47.7	47.7	65.8	57.7
Product C	45.3	55.4	45.4	45.3
Product D	52.4	30.9	32.8	44.4
Product E	39.9	53.1	40.9	35.9

Visualization (Easy to see trends)



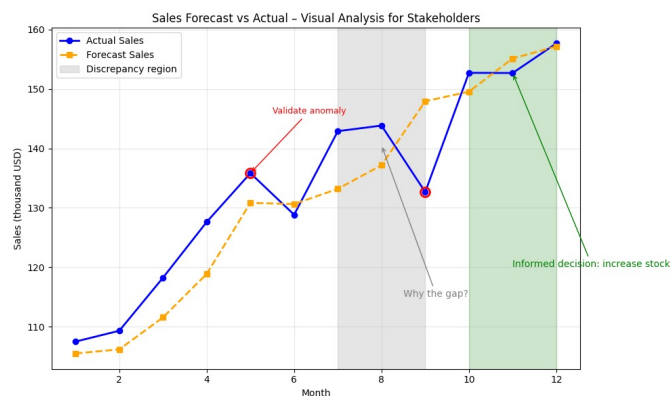
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Interpreting Model Outputs through Visualization

Enables stakeholders to:

- Validate results
- Ask meaningful questions
- Make informed decisions based on visual evidence



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Role of visualization

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- Support interpretation of analytics and machine learning results
- Enable comparison across models, parameters, and time periods
- Reveal patterns, trends, and anomalies in large-scale data
- Facilitate communication between analysts and stakeholders
- Improve decision-making through clear and evidence-based insights

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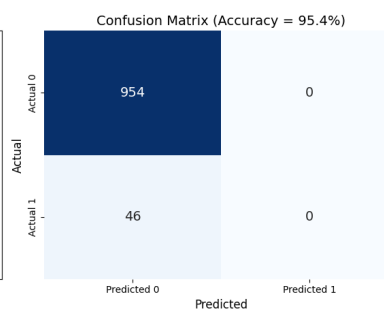
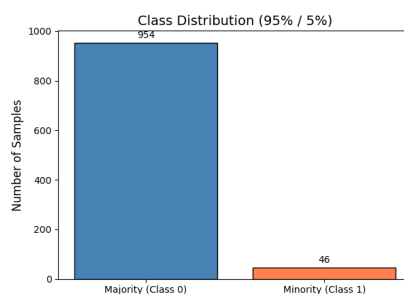
When 95% Accuracy Still Fails

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Problem: Imbalanced Data

- Dataset dominated by one class (e.g., 95% non-fraud, 5% fraud)
- Model predicts the majority class → high accuracy
- Minority class is poorly detected

Problem: Imbalanced Dataset



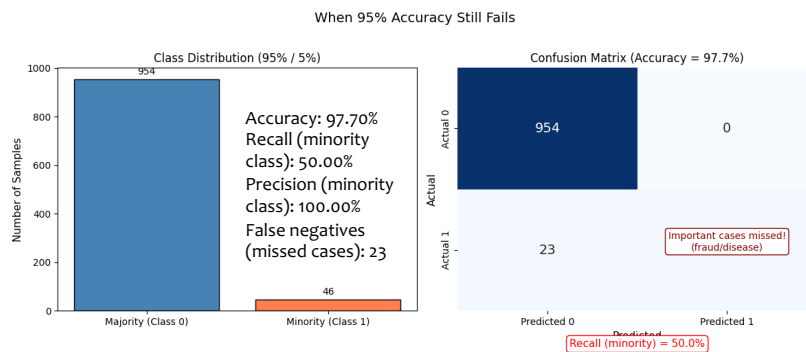
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When 95% Accuracy Still Fails

What Really Happens

- Accuracy looks high, but recall is low
- Important cases are missed (e.g., fraud, disease)
- Confusion matrix reveals hidden errors



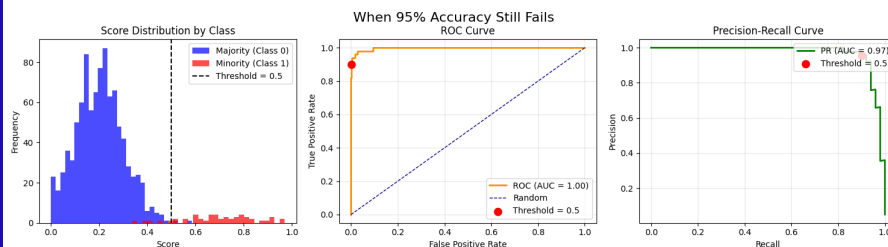
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When 95% Accuracy Still Fails

Why Accuracy is Misleading

- Precision & Recall show true performance
- Threshold selection affects classification results
- ROC / PR curves help evaluate trade-offs



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Challenges when visualizing Big Data results

Data scale and volume

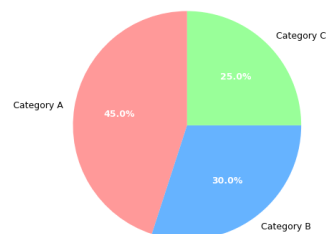
- Large datasets cannot be visualized point-by-point
- Requires aggregation, sampling, or summarization

High dimensionality

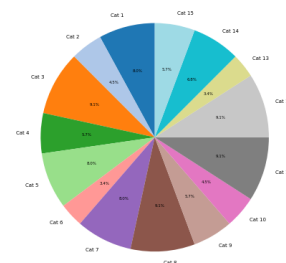
- Many features make direct visualization difficult
- Often needs dimensionality reduction or projection

No bad chart by default — it depends on context

Pie chart with 3 categories
✓ Good for showing proportions



Pie chart with 15 categories
✗ Overwhelming, hard to interpret



A chart becomes bad when used in the wrong context or for the wrong audience.

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Challenges when visualizing Big Data results

Information overload

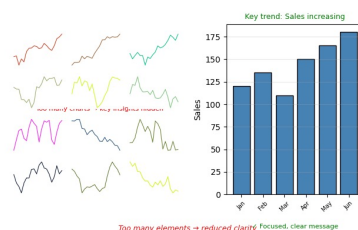
- Too many charts or visual elements reduce clarity
- Key insights may be hidden in cluttered visuals

Performance and interactivity

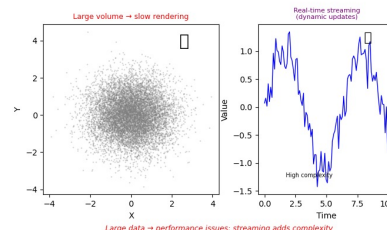
- Large data volumes can slow down rendering and interaction
- Real-time or streaming data increases complexity

Challenges when Visualizing Big Data Results

Information Overload



Performance and Interactivity



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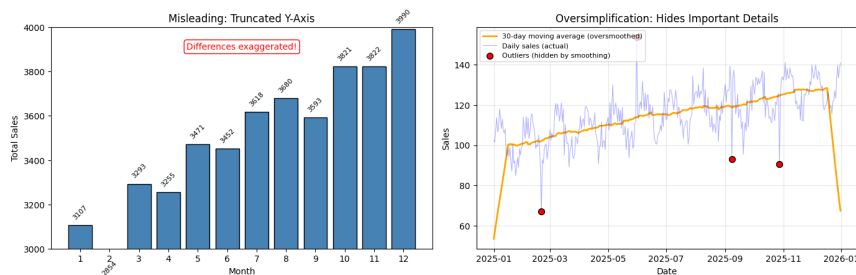


Challenges when visualizing Big Data results

Interpretability risks

- Poor visualization choices can mislead users
- Oversimplification may hide important details

Challenges when Visualizing Big Data Results



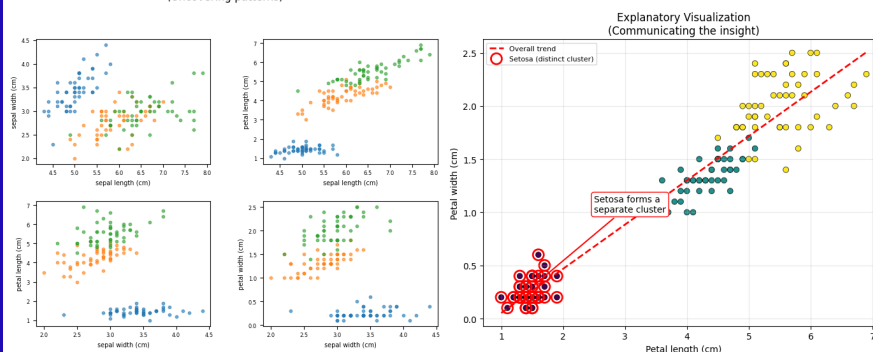
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Exploratory vs. Explanatory Visualization

- Exploratory visualization helps you uncover insights.
- Explanatory visualization helps you communicate the insight.

Exploratory Visualization (Uncovering patterns) Exploratory vs. Explanatory Visualization



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Exploratory vs. Explanatory Visualization

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Exploratory Visualization	Explanatory Visualization
Explore data and discover patterns	Communicate insights and conclusions
Used by analysts and data scientists	Used for decision makers and stakeholders
Flexible and iterative	Focused and carefully designed
May be unpolished or temporary	Clean and presentation-ready
Many views and variations	Selected key views only
Answers: "What is happening in the data?"	Answers: "What should we know or do?"

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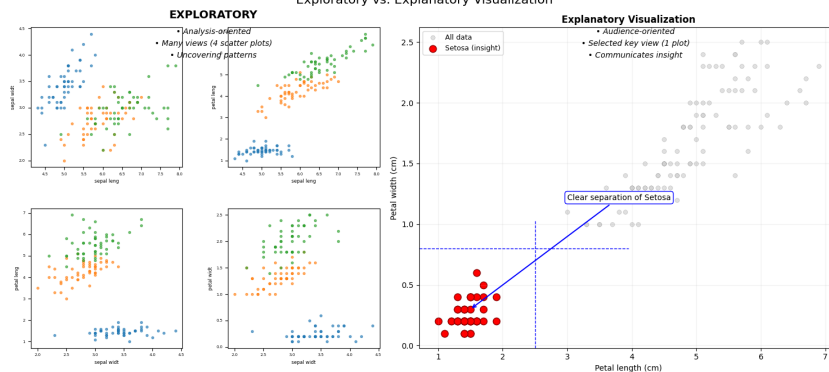
Exploratory vs. Explanatory Visualization

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Key Differences

- Exploration vs. communication
- Analysis-oriented vs. audience-oriented
- Many views vs. selected key views

Exploratory vs. Explanatory Visualization



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Example

- Dataset:** Customer transaction data (sales, time, customer segments)



Source: devopedia

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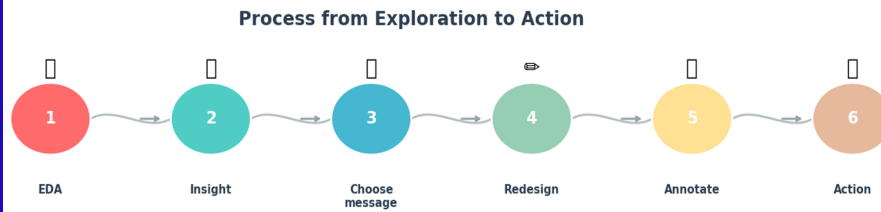
Example

Exploratory Visualization (Before)	Explanatory Visualization (After)
Multiple charts showing sales by time, region, and customer type	One focused chart showing key sales trend
Interactive filters and trial-and-error views	Fixed view highlighting main insight
Rough layouts, quick plots	Clean design with titles and annotations
Used to explore patterns and anomalies	Used to support a business decision
Goal: understand the data	Goal: explain the insight

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Visualization Workflow



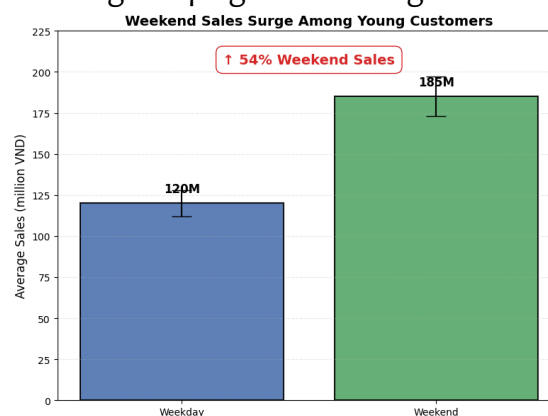
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Example

Key Insight (After):

- Weekend sales increased significantly among young customers → marketing campaign should target weekends.



Key Insight: Weekend sales increased significantly → target weekends for marketing

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Visualizations for Models and Metrics

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- Focus on communicating model results, not model implementation
- Choose visualizations that clearly show:
 - Model performance
 - Differences between models or scenarios
 - Key patterns or outcomes
- Prefer simple visuals for communication
- Avoid technical or overloaded charts

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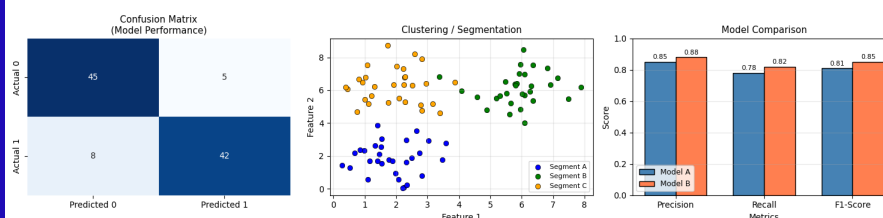


Visualizations for Models and Metrics

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- Examples of common choices:
 - Model performance → simple charts (e.g., confusion matrix, trend lines)
 - Grouping or segmentation → clear cluster or category visuals
 - Model comparison → side-by-side or summary charts

Visualizations for Models and Metrics

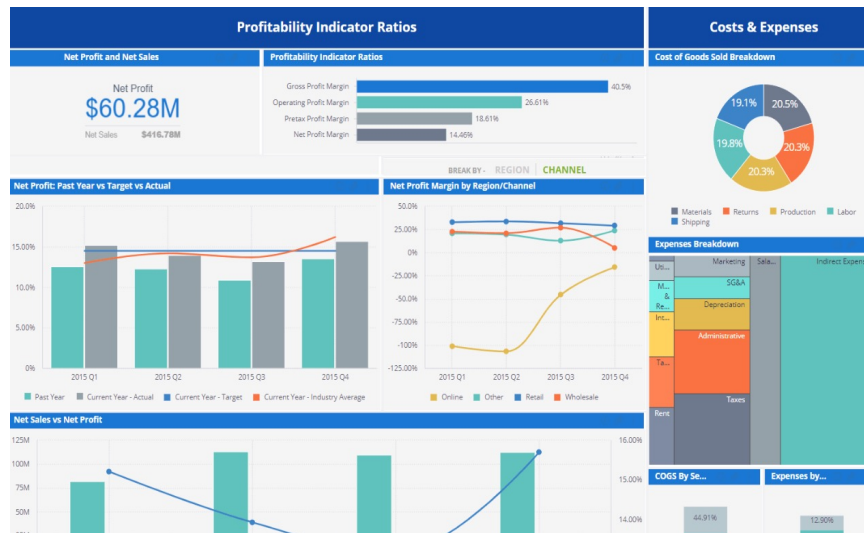


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Visualizations for Models and Metrics

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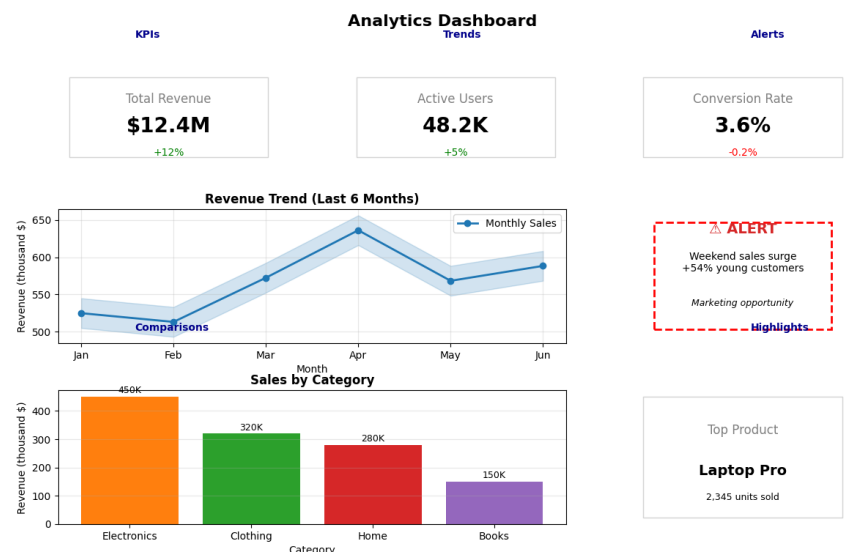
Source: sisense

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Visualizations for Models and Metrics

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Dashboards for Analytics Communication

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- A dashboard is a single screen that summarizes key analytics results
- Used to communicate insights, not to show all raw data
- Typically includes:
 - Key performance indicators (KPIs)
 - Important trends and comparisons
 - Alerts or highlights

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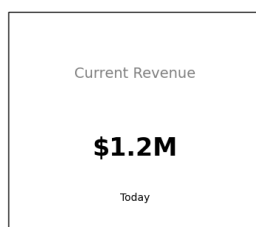
Dashboards for Analytics Communication

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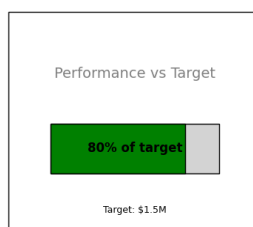
- Designed for non-technical users and decision makers
- Helps users quickly answer:
 - What is happening now?
 - Is the performance good or bad?
 - What needs attention?

Dashboards for Analytics Communication

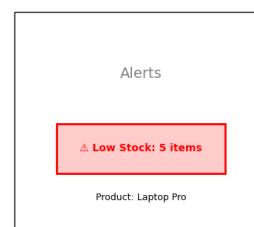
What is happening now?



Is the performance good or bad?



What needs attention?



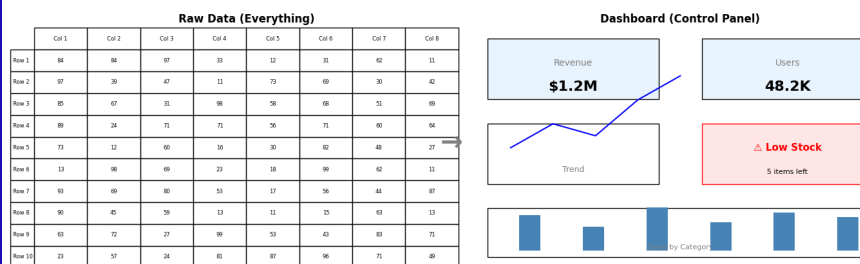
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Dashboards for Analytics Communication

- A dashboard is like a “control panel” for analytics results you don’t see everything, only what matters for decisions.

Dashboards for Analytics Communication



A dashboard is like a “control panel” – you don’t see everything, only what matters for decisions.

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Data Storytelling for Decision Making

- Combine data, visualization, and narrative to convey insights
- Focus on why the result matters, not just what the result is
- Structure a data story:
 - Context: the problem or question
 - Insight: key finding from analytics
 - Evidence: supporting visualizations
 - Action: recommended decision

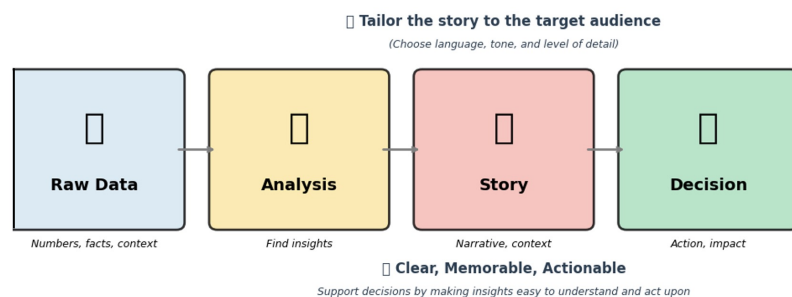
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Data Storytelling for Decision Making

- Tailor the story to the target audience
- Support decisions by making insights clear, memorable, and actionable
- Data storytelling turns charts into messages that support

Data Storytelling for Decision Making



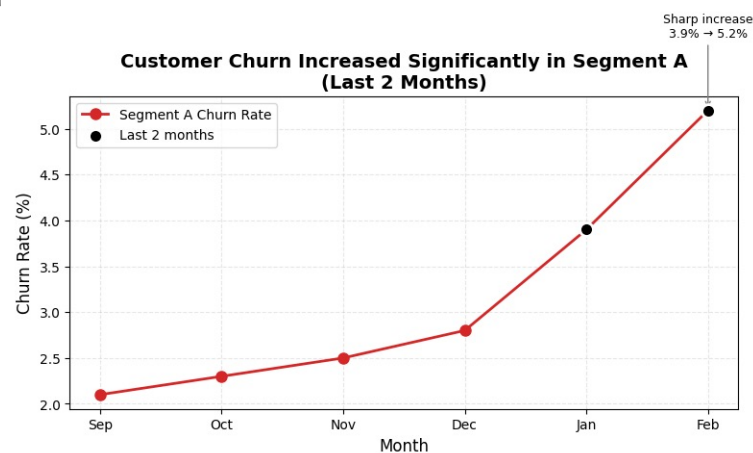
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Mini Data Story Example

Context

- Customer churn increased significantly in Segment A over the past 2 months.



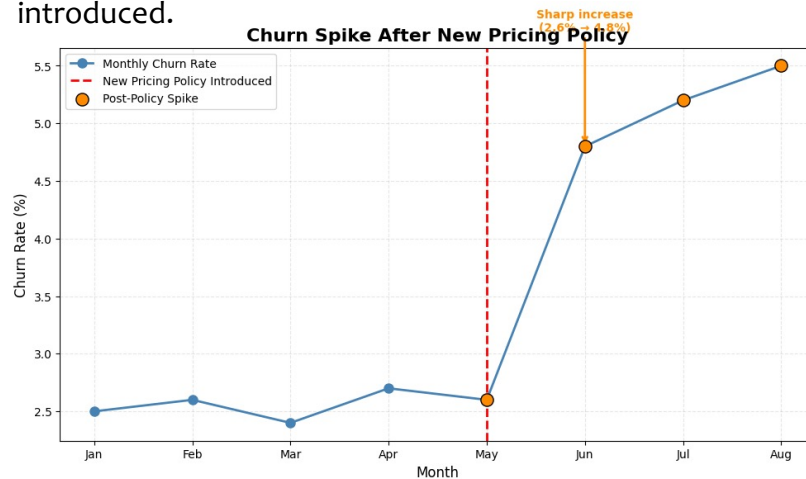
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Mini Data Story Example

Insight

- Churn spiked immediately after the new pricing policy was introduced.



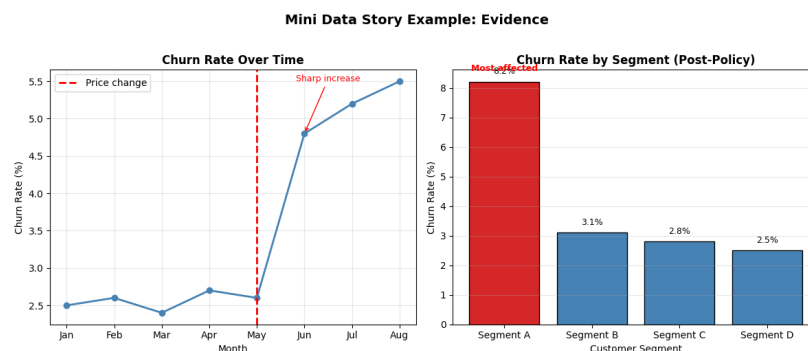
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Mini Data Story Example

Evidence

- The line chart shows the churn rate rising after the price change
- The segment bar chart shows that Segment A is the most affected.



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Mini Data Story Example

Action

- Review pricing changes for Segment A
- Offer retention incentives to affected customers
- Monitor churn trend after adjustments

Mini Data Story Example: Actions



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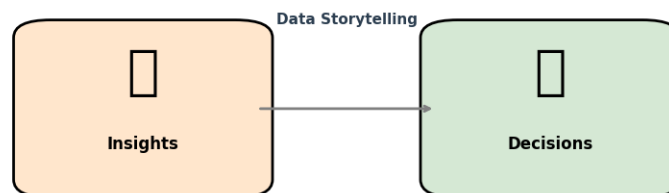


Mini Data Story Example

Key Message

- Data storytelling turns insights into decisions.

Mini Data Story Example



Key Message

Data storytelling turns insights into decisions.

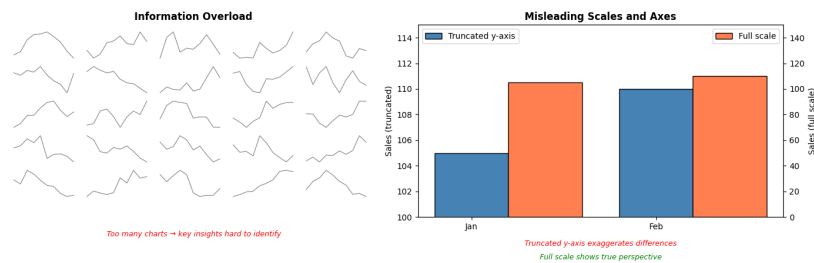
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Common Visualization Pitfalls in Big Data

- Information overload
 - Too many charts or metrics on one screen
 - Key insights become hard to identify
- Misleading scales and axes
 - Truncated or inconsistent axes distort interpretation
 - Visual differences may be exaggerated or hidden

Common Visualization Pitfalls in Big Data



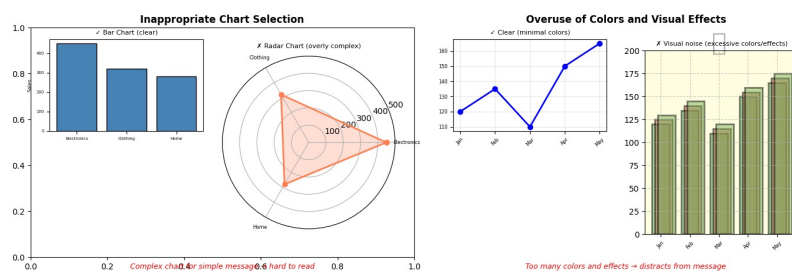
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Common Visualization Pitfalls in Big Data

- Inappropriate chart selection
 - Using complex charts for simple messages
 - Choosing charts that do not match data or purpose
- Overuse of colors and visual effects
 - Excessive colors, 3D effects, or animations reduce clarity
 - Visual noise distracts from the message

Common Visualization Pitfalls in Big Data



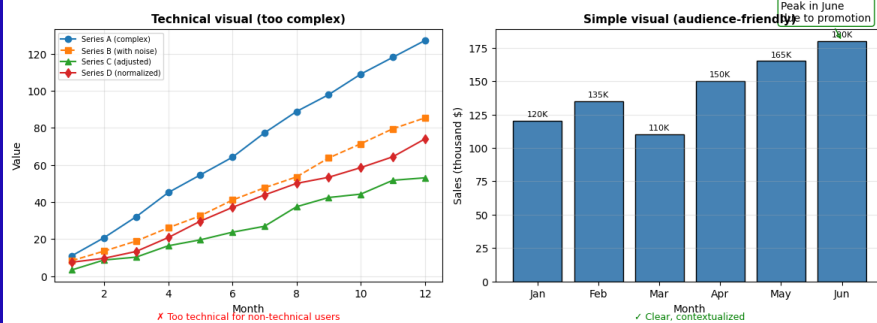
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Common Visualization Pitfalls in Big Data

- Ignoring audience needs
 - Technical visuals shown to non-technical users
 - Missing context or explanation

Ignoring Audience Needs



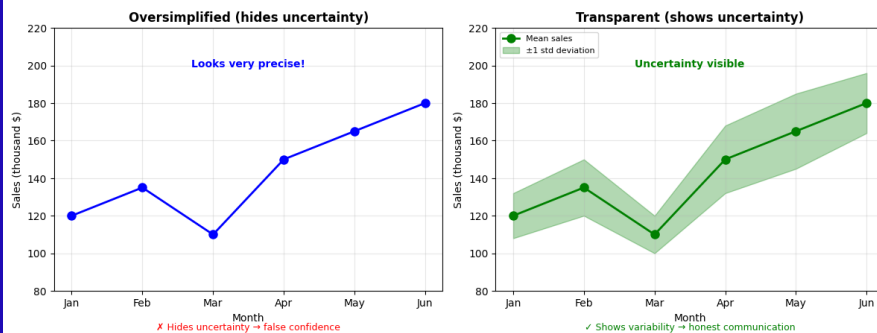
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Common Visualization Pitfalls in Big Data

- Oversimplification of Big Data results
 - Hiding uncertainty, bias, or limitations
 - Creating false confidence in analytics outcomes

Oversimplification of Big Data Results



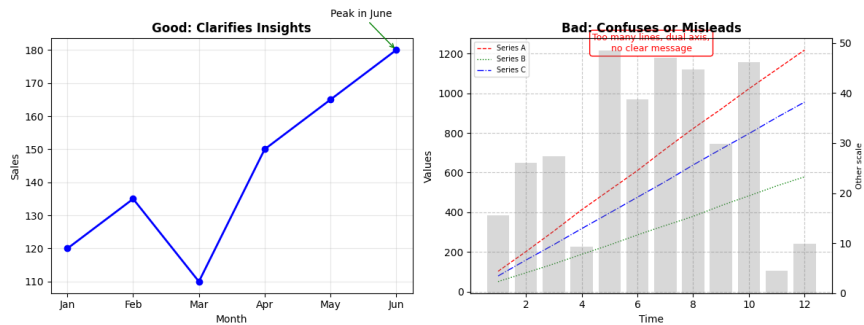
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Common Visualization Pitfalls in Big Data

- A good visualization should clarify insights, not confuse or mislead.
- A visualization pitfall is a mistake that makes data harder to understand or easy to misinterpret.

Visualization Pitfalls: Clarify vs. Confuse



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Example



Top 3 Good and Bad Examples of Data Visualizations

Source: Chartexpo

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Example

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Box Plot (Top-Left)

Good

- Shows data distribution (median, quartiles, outliers)
- Effective for comparing multiple groups
- Useful for analytical exploration

Bad (when misused)

- Difficult for non-technical audiences
- May confuse stakeholders unfamiliar with statistical charts

Lesson:

- A technically correct visualization can still fail if the audience cannot interpret it.

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Example

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Stacked Bar + Line Chart (Bottom-Left)

Good

- Combines composition (bars) and trend (line)
- Useful for monitoring performance over time

Bad (when overloaded)

- Too many colors and elements
- Hard to identify the main message

Lesson:

- Combining charts is effective only when the focus is clear.

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Example

Sankey Diagram (Right)

Good

- Clearly represents flows and transitions
- Powerful for showing movement and proportions

Bad (for large or complex data)

- Becomes cluttered with many flows
- Difficult to extract a single key insight

Lesson:

- Powerful visualizations must be simplified to remain interpretable.

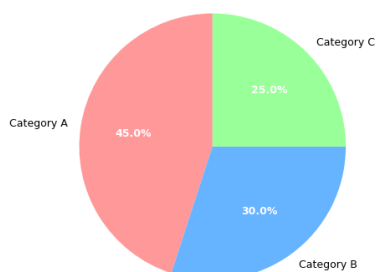


Key Takeaway

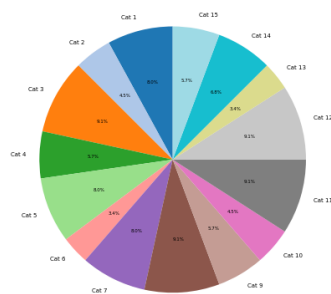
- There is no bad chart by default.
A chart becomes bad when it is used in the wrong context or for the wrong audience.

No bad chart by default — it depends on context

Pie chart with 3 categories
✓ Good for showing proportions



Pie chart with 15 categories
✗ Overwhelming, hard to interpret



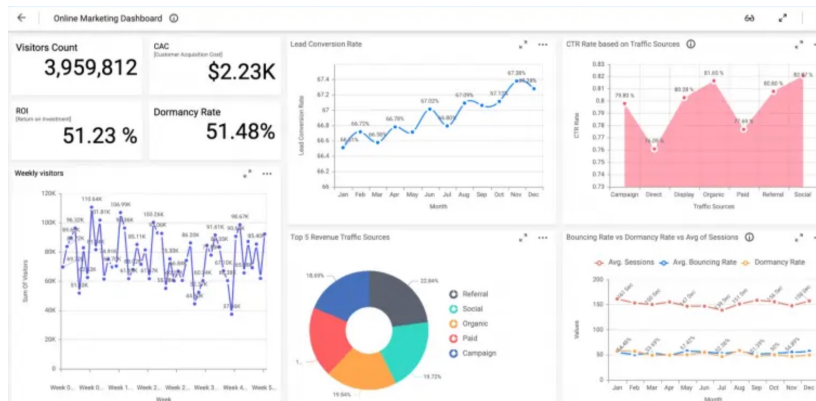
A chart becomes bad when used in the wrong context or for the wrong audience.



Applications & Case Studies

Business Analytics

- Dashboards to monitor KPIs, sales trends, and customer behavior
- Visualization supports managerial decision making



Source: boldbi

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Applications & Case Studies

Healthcare Analytics

- Visualizing patient risk, outcomes, and trends
- Helps clinicians understand analytics results quickly



Source: demigos

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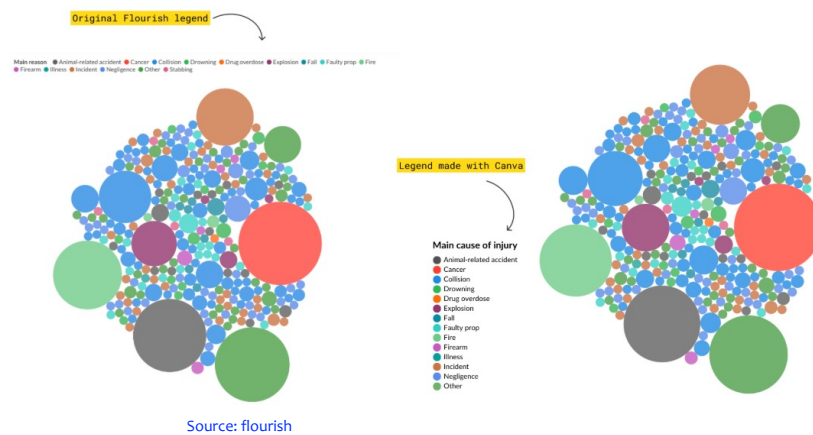


Applications & Case Studies

Social Media & Marketing Analytics

- Visualizing engagement, sentiment, and user segments
- Supports campaign evaluation and strategy adjustment

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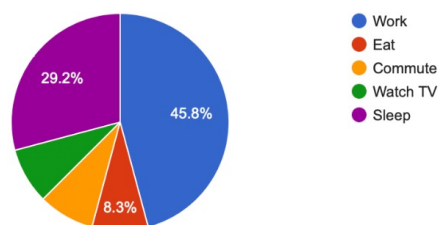
Applications & Case Studies

Smart City & IoT Analytics

- Visualizing traffic, sensor data, and real-time events
- Enables monitoring and timely interventions

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Pie Chart - [view source](#)



[more](#)

Source: sproutsocial

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Thank you for your attention!